CELLULAR AGING · VIVIX+™

# The Zombie Cell Problem

## Why aging seems to speed up in our later years, and what your cells need to push back.

Shared by Ann and Benjamin

Almost everyone over 65 knows the feeling. A cold that once lasted three days now lingers for three weeks. A bruise takes a month to fade. Stairs that were nothing are now something. It feels like aging picked up speed.

It did, and scientists now understand why.

This piece was prompted by the widely shared video ***The REAL Reason Your Body Ages FASTER After 75*** from the Dr. Aven Royce channel on YouTube, viewed nearly 800,000 times. Its main point: aging does not run in a straight line. It speeds up when damaged "zombie cells" build up faster than an aging immune system can clear them. Credit to the channel for explaining it so clearly. Below, we tie that idea to the published research, and to what Vivix+™ does about it.

## Cells that stop working but won't die

Every cell can divide only so many times. When a cell reaches that limit, or is damaged along the way, it usually does not die. It enters a state called senescence. Researchers nicknamed these cells zombie cells.

A zombie cell stops doing its job but stays in your tissue. Worse, it sends out a steady stream of inflammatory signals. Those signals damage the healthy tissue around it and can push neighboring cells to become zombie cells too.

This matters. In a landmark 2011 study, clearing zombie cells from aging mice delayed age-related decline in their muscles, fat, and eyes.[1] Mayo Clinic has since shown that the number of zombie cells can be reduced in people.[2]

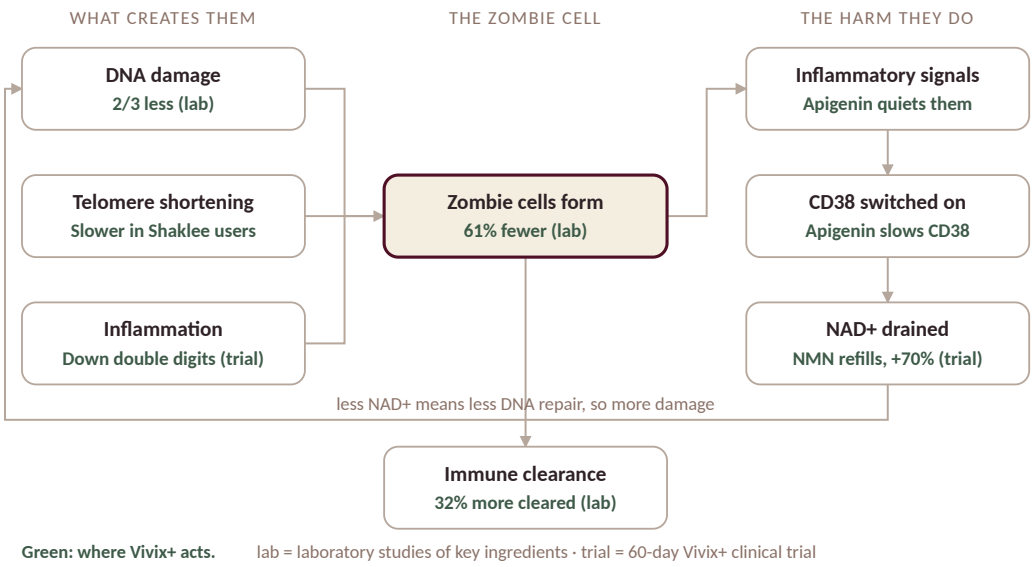
## Why they pile up

Zombie cells build up when they form faster than the body clears them. Both sides of that balance shift as we age.

**More of them form.** DNA damage, shortening telomeres (the protective caps on our chromosomes), and inflammation all push cells into senescence. Nobel laureate Elizabeth Blackburn and psychologist Elissa Epel showed that chronic stress speeds up telomere shortening.[3]

**Fewer get cleared.** Our immune system normally removes zombie cells. As it ages, it falls behind, and zombie cells accumulate faster.[4]

**And they drain your energy.** In 2020, teams at the Buck Institute and Mayo Clinic discovered that zombie cells switch on an enzyme called CD38 in nearby immune cells. CD38 destroys NAD+, the molecule your cells need to make energy and repair DNA.[5,6] Less NAD+ means less repair, which means more damage. The loop feeds itself.




The zombie-cell loop, and the five places Vivix+™ acts on it.

## Where Vivix+™ fits

Vivix+™ was designed to work on both halves of this problem: the damage that creates zombie cells, and the NAD+ they drain away.

### Fewer zombie cells formed, more cleared LABORATORY

In Shaklee's laboratory studies, key ingredients in Vivix+™ led to 61% fewer zombie cells forming and 32% more being cleared.

### Quieter signals PUBLISHED RESEARCH

Apigenin, an ingredient in Vivix+™, suppressed the inflammatory signals of zombie cells in research from Judith Campisi's lab at the Buck Institute.[7]

### NAD+ refilled and protected CLINICAL + RESEARCH

Vivix+™ raises NAD+ with NMN, and apigenin slows CD38, the enzyme that zombie cells switch on to destroy NAD+.[8] In the clinical trial, NAD+ rose 70% within 24 hours.

### Less damage at the source CLINICAL + LABORATORY

The polyphenol blend at the heart of Vivix, including muscadine grape and resveratrol, blunted the inflammatory spike after a fast-food meal in a published clinical study.[9] In laboratory studies, key ingredients cut DNA damage by two thirds.

## What happened in people

In a randomized, placebo-controlled trial, healthy adults aged 45 to 65 took Vivix+™ or a placebo for 60 days.

| +3.2 | Double digits | 70% |
|---|---|---|
| $VO_2$ max points, about 8 years of age-related decline, reversed | Drop in key inflammatory markers | More NAD+ in 24 hours |

$VO_2$ max, how well your body turns oxygen into energy, is one of the strongest predictors of healthy, active years. It usually rises only with hard training. Here it rose without it.

The trial found one more thing. People taking Vivix+™ reported feeling more connected to the people around them. The video makes a point of this too: social connection is one of the strongest supports for healthy aging.

Long-term users of Shaklee supplements, Vivix among them, also had longer telomeres than healthy peers in two studies, with telomere shortening slowed by as much as 57%.[10,11]

## What this means for you

Zombie cells are not a fixed sentence. The video's closing point is the right one: the years ahead are still being written by the choices we make now. Strength training, good sleep, real food, managing stress, and staying connected all work on the same loop.

Vivix+™ adds support at the level where it all begins, inside your cells. It helps you make fewer zombie cells, calm the signals they send, and keep the NAD+ your cells run on.

### Start with Vivix+™

Order two bottles and you become a Shaklee member automatically: $96 a bottle, free shipping, and a 100% money-back guarantee.

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Vivix+™ trial and laboratory results: Shaklee Vivix+™ product information.


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These statements have not been evaluated by the Food and Drug Administration. This product is not intended to diagnose, treat, cure, or prevent any disease. Laboratory results describe studies of Vivix+ key ingredients in cells, not results in people.